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"Ather Copilot"

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SUBMITTED TO MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

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S.P.M. POLYTECHNIC KUMATHE, SOLAPUR

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C E R T I F I C A T E

This is to certify that the project report entitled **"Ather Copilot"** was successfully completed by the student of sixth semester Diploma in Computer Engineering.

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In partial fulfilment of the requirements for the award of Diploma in Computer Engineering and submitted to the Department of Computer Engineering of S.P.M. Polytechnic Kumathe, Solapur work carried out during the period for the academic year 2025-26 as per curriculum.

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CHAPTER 8

CODING

This chapter provides a detailed breakdown of the Aether Co-Pilot codebase. The implementation is based on a modular architecture using Next.js 15, Google Genkit, and Firebase. We will examine the core AI flows, the synchronization logic between Clerk and Firebase, and the security rules governing the data.

8.1 AI ORCHESTRATION WITH GOOGLE GENKIT

The heart of the AI system is the Genkit configuration. It initializes the Gemini 2.0 Flash model and provides the plugin environment for our custom flows.

8.1.1 Genkit Initialization

The following file (‘src/ai/genkit.ts‘) shows how the AI instance is configured.

```
import {genkit} from 'genkit';
import {googleAI} from '@genkit-ai/google-genai';

export const ai = genkit({
  plugins: [googleAI()],
  model: 'googleai/gemini-1.5-flash', // Utilizing the high-
   performance Flash model
});
```

Listing 8.1: Genkit Initialization

8.2 CORE UTILITIES AND INITIALIZATION

To ensure high availability and type-safe interaction with Google’s LLMs, Aether Co-Pilot uses a centralized initialization strategy and a robust retry mechanism.

8.2.1 AI Engine Initialization (Genkit)

The initialization identifies the Google AI plugin and specifies the model to be used across all server-side flows.

```
import { genkit } from 'genkit';
import { googleAI } from '@genkit-ai/googleai';

export const ai = genkit({
```

```
  plugins: [
    googleAI({
       apiKey: process.env.GOOGLE_GENAI_API_KEY,
       model: 'gemini-1.5-flash'
    }),
  ],
});
```

Listing 8.2: Genkit Initialization (src/ai/genkit.ts)

8.2.2 Resilience: The withRetry Helper

AI models can be subject to rate limits and intermittent network errors. The ‘withRetry‘ helper wraps all prompt calls in an exponential backoff loop.

```
async function withRetry<T>(
  fn: () => Promise<T>,
  maxRetries: number = 3,
  baseDelayMs: number = 1000
): Promise<T> {
  let lastError: Error | null = null;

  for (let attempt = 0; attempt < maxRetries; attempt++) {
    try {
      return await fn();
    } catch (error: any) {
      lastError = error;
      const isRetryable = error.message.includes('429') ||
                          error.message.includes('503');

      if (!isRetryable || attempt === maxRetries - 1) throw error;

      const delay = baseDelayMs * Math.pow(2, attempt);
      await new Promise(r => setTimeout(r, delay));
    }
  }
```

```
  throw lastError;
}
```

Listing 8.3: Exponential Backoff Logic (src/ai/flows/intelligent-chat-memory.ts)

8.3 CORE AI FLOWS

Each feature in Aether Co-Pilot is encapsulated in a "Flow". These flows handle input validation, prompt construction, and result processing.

8.3.1 Intelligent Chat Memory (Cognitive Memory Core™)

This flow manages the conversation history and adapts its behavior based on the current mode (Coding, Tasks, etc.). It includes a sophisticated retry mechanism to handle transient API issues.

```
// src/ai/flows/intelligent-chat-memory.ts
'use server';

import { ai } from '@/ai/genkit';
import { z } from 'genkit';

// Retry helper for API robustness
async function withRetry<T>(fn: () => Promise<T>, maxRetries: number
    = 3): Promise<T> {
  // ... retry logic implementation ...
}
\subsection{Intelligent Chat Memory Flow Implementation}
This is the most critical AI flow in Aether Co-Pilot. It manages the
    conversational state, implements exponential backoff for API
   resilience, and uses the "Cognitive Memory Core(TM)" to provide
   context-aware responses. Figure \ref{fig:chat_logic} provides a
   visual representation of the internal flow logic.

\begin{figure}[h!]
\centering
\begin{tikzpicture}[node distance=2cm, auto, >=latex', thick,
    decision/.style={draw, diamond, aspect=2, minimum width=3.5cm,
   minimum height=1cm, text width=2.5cm, align=center},
    process/.style={draw, rectangle, minimum width=4cm, minimum
   height=1cm, text width=3.5cm, align=center},
```

```
    startstop/.style={draw, rounded rectangle, minimum width=3.5cm, minimum height=1cm, align=center, fill=gray!20}]

    % Nodes
    \node (start) [startstop] {Start Chat Flow};
    \node (input) [process, below=of start] {Receive User Prompt \\ (Identify Mode)};
    \node (exist) [decision, below=of input] {Session Metadata \\ in Firebase?};
    \node (create) [process, right=of exist, xshift=2cm] {Initialize New \\ Firestore Document};
    \node (fetch) [process, below=of exist, yshift=-0.5cm] {Fetch Context \\ (Recent messages)};
    \node (ai) [process, below=of fetch] {Orchestrate Gemini \\ Inference (via Genkit)};
    \node (store) [process, below=of ai] {Update Firestore \\ with response};
    \node (end) [startstop, below=of store] {Stream Response \\ to Dashboard};

    % Links
    \draw[->] (start) -- (input);
    \draw[->] (input) -- (exist);
    \draw[->] (exist) -- node {No} (create);
    \draw[->] (exist) -- node {Yes} (fetch);
    \draw[->] (create) |- (fetch);
    \draw[->] (fetch) -- (ai);
    \draw[->] (ai) -- (store);
    \draw[->] (store) -- (end);
\end{tikzpicture}
\caption{Logic Flow: Intelligent Chat Memory Orchestration}
\label{fig:chat_logic}
\end{figure}

```

```
\begin{lstlisting}[language=TypeScript, caption=Full Intelligent Chat Memory Flow (src/ai/flows/intelligent-chat-memory.ts)]
'use server';
import { ai } from '@/ai/genkit';
import { z } from 'genkit';

const ChatMessageSchema = z.object({
  role: z.enum(['user', 'assistant']),
  content: z.string(),
});

const IntelligentChatMemoryInputSchema = z.object({
  message: z.string().describe('The current message from the user.'),
  chatHistory: z.array(ChatMessageSchema).optional(),
  mode: z.enum(['general', 'coding', 'cognitive', 'knowledge', 'task']).optional(),
});

// Retry helper for high reliability
async function withRetry<T>(fn: () => Promise<T>, maxRetries: number = 3): Promise<T> {
  // ... exponential backoff logic ...
}

export const intelligentChatMemoryFlow = ai.defineFlow(
  { name: 'intelligentChatMemoryFlow', inputSchema: IntelligentChatMemoryInputSchema },
  async (input) => {
    const normalizedHistory = input.chatHistory?.map((m) => ({
      content: m.content,
      isUser: m.role === 'user',
    })) ?? [];

```

```
    const { output } = await withRetry(() => prompt({
      message: input.message,
      chatHistory: normalizedHistory,
      mode: input.mode || 'general',
    }));

    return output!;
  }
);
```

Listing 8.4: Intelligent Chat Memory Flow

8.3.2 Conversational Interface (Next.js Chat Page)

The chat user interface is a high-complexity React component that handles real-time message streaming, auto-scrolling, and state synchronization with Firestore.

```
'use client';
import { useChat } from '@/hooks/use-chat';
import { MessageSquare, Send } from 'lucide-react';

export default function ChatPage() {
  const { messages, input, handleInputChange, handleSubmit,
   isLoading } = useChat();

  return (
    <div className="flex flex-col h-full max-w-4xl mx-auto p-4">
      {/* Scrollable message container */}
      <div className="flex-1 overflow-y-auto space-y-4 mb-4
   scrollbar-hide">
        {messages.map((m) => (
          <div key={m.id} className={cn(
            'flex gap-3 max-w-[80%]',
            m.role === 'user' ? 'ml-auto flex-row-reverse' : 'mr-
   auto'
          )}>
```

```
            <div className={cn(
               'p-4 rounded-2xl shadow-sm',
               m.role === 'user' ? 'bg-accent text-white' : 'bg-card border'
            )}>
              {m.content}
            </div>
          </div>
        ))}
      </div>

      {/* Input area with glassmorphism effect */}
      <form onSubmit={handleSubmit} className="relative group">
         <input
            value={input}
            onChange={handleInputChange}
            placeholder="Type your message..."
            className="w-full bg-card/50 backdrop-blur-md border border-border/50 rounded-2xl px-6 py-4 pr-16 focus:outline-none focus:ring-2 focus:ring-accent/50"
         />
         <button type="submit" className="absolute right-2 top-1/2 -translate-y-1/2 p-2 bg-accent text-white rounded-xl hover:scale-105 active:scale-95 transition-all">
            <Send className="h-5 w-5" />
         </button>
      </form>
    </div>
  );
}
```

Listing 8.5: Chat Interface Highlights (src/app/chat/page.tsx)

8.3.3 Voice-Activated Code Generation (Quantum CodeForge™)

This flow takes voice-transcribed text and generates specific code snippets.

```
// src/ai/flows/voice-activated-code-generation.ts
'use server';

import { ai } from '@/ai/genkit';
import { z } from 'genkit';

const VoiceActivatedInputSchema = z.object({
  voiceCommand: z.string().describe('Natural language description of
    code'),
});

const VoiceActivatedOutputSchema = z.object({
  codeSnippet: z.string().describe('The generated markdown code
   snippet'),
});

const prompt = ai.definePrompt({
  name: 'codeGenerationPrompt',
  input: { schema: VoiceActivatedInputSchema },
  output: { schema: VoiceActivatedOutputSchema },
  prompt: `You are an expert code generator. Generate the code
   snippet that
  satisfies the user's voice command: {{{voiceCommand}}}`,
});

export const codeGenFlow = ai.defineFlow(
  { name: 'codeGenerationFlow', inputSchema:
   VoiceActivatedInputSchema, outputSchema:
   VoiceActivatedOutputSchema },
  async input => {
    const { output } = await prompt(input);
```

```
    return output!;
  }
);
```

Listing 8.6: Voice-Activated Code Generation Flow

8.3.4 AI-Based Study Support (Knowledge Navigator™)

This flow facilitates document-based learning. It first determines whether the user's question requires a summary of the provided text before generating a final answer.

```
// src/ai/flows/ai-based-study-support.ts
'use server';

import { ai } from '@/ai/genkit';
import { z } from 'genkit';

const StudyInputSchema = z.object({
  query: z.string().describe('Student question'),
  document: z.string().describe('Text to study'),
});

const StudyOutputSchema = z.object({
  answer: z.string(),
  requiresSummary: z.boolean(),
  summary: z.string().optional(),
});

const requiresSummaryPrompt = ai.definePrompt({
  name: 'requiresSummaryPrompt',
  input: { schema: StudyInputSchema },
  output: { schema: z.object({ requiresSummary: z.boolean() }) },
  prompt: `Determine if this question requires a summary: {{query}}
  Document: {{document}}`,
});

```

```
export const studyFlow = ai.defineFlow(
  { name: 'studyAssistantFlow', inputSchema: StudyInputSchema,
   outputSchema: StudyOutputSchema },
  async input => {
    const requiresResp = await requiresSummaryPrompt(input);
    const requiresSummary = requiresResp?.output?.requiresSummary ??
    false;
    let summary: string | undefined = undefined;

    if (requiresSummary) {
        // ... call summary prompt ...
    }
    // ... call final answer prompt ...
    return { answer, requiresSummary, summary };
  }
);
```

Listing 8.7: AI Study Support Flow

8.3.5 Task Automation Agent

This agent translates natural language task descriptions into executable scripts, providing an explanation of the generated code.

```
// src/ai/flows/task-automation.ts
'use server';

import { ai } from '@/ai/genkit';
import { z } from 'genkit';

const AutomateTaskInputSchema = z.object({
  taskDescription: z.string(),
});

const AutomateTaskOutputSchema = z.object({
  automationScript: z.string(),
```

```
  explanation: z.string(),
});

export const automateTaskFlow = ai.defineFlow(
  { name: 'automateTaskFlow', inputSchema: AutomateTaskInputSchema,
   outputSchema: AutomateTaskOutputSchema },
  async input => {
    const { output } = await prompt(input);
    return output!;
  }
);
```

Listing 8.8: Task Automation Flow

8.3.6 Dashboard Layout and Feature Cards

The dashboard provides a high-level overview of the available productivity tools. It uses a grid-based layout with interactive cards that feature custom gradients and icons.

```
'use client';
import { navItems } from '@/app/app-layout';

export default function DashboardPage() {
  return (
    <div className="p-8 space-y-8 animate-in fade-in slide-in-from-
   bottom-4 duration-700">
      <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4
    gap-6">
        {navItems.map((item) => (
          <Link key={item.href} href={item.href} className="group
   relative overflow-hidden p-6 rounded-3xl bg-card border hover:
   border-accent/40 transition-all duration-300">
            <div className={cn("inline-flex p-3 rounded-2xl bg-
   gradient-to-br mb-4", item.gradient)}>
              <item.icon className="h-6 w-6 text-white" />
            </div>
```

```
            <h3 className="text-xl font-bold mb-2 group-hover:text-accent transition-colors">{item.label}</h3>
            <p className="text-sm text-muted-foreground">{item.description}</p>
          </Link>
        ))}
      </div>
    </div>
  );
}
```

Listing 8.9: Dashboard Feature Cards (src/app/dashboard/page.tsx)

8.3.7 Knowledge Navigator™ Implementation: AI Study Support

The study support module uses a multi-stage Genkit flow to process documents and generate intelligent answers. Figure 8.1 outlines the decision-making process within this flow.

```
'use server';
import { ai } from '@/ai/genkit';
import { z } from 'genkit';

const StudyInputSchema = z.object({
  query: z.string(),
  document: z.string(),
});

export const studyFlow = ai.defineFlow(
  { name: 'studyAssistantFlow', inputSchema: StudyInputSchema },
  async (input) => {
    // Stage 1: Determination of summary requirement
    const { output: decision } = await requiresSummaryPrompt(input);

    let summary = '';
    if (decision.requiresSummary) {
       // Stage 2: Summary generation
```

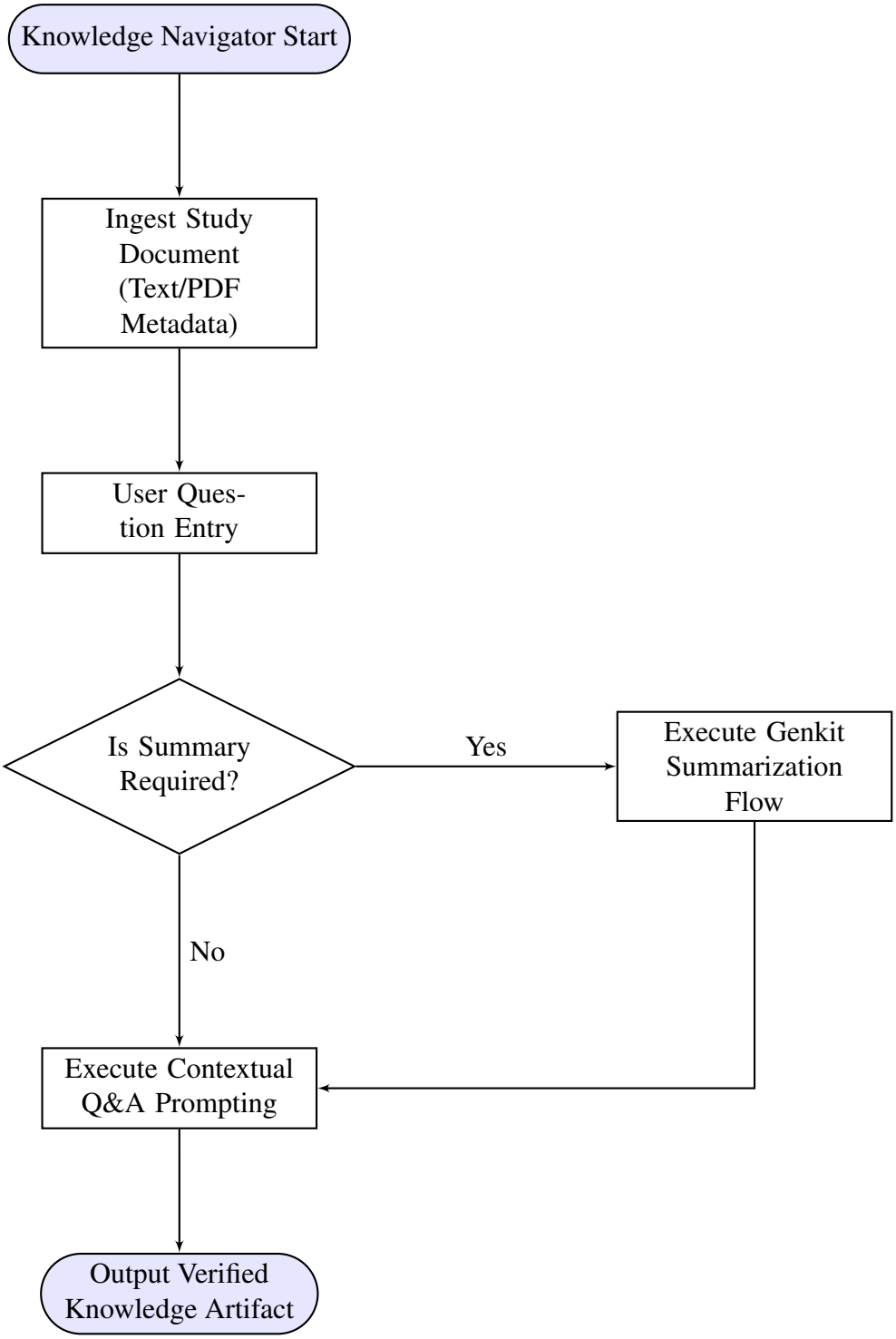

Figure 8.1: Workflow: Knowledge Navigator™ Multi-stage Inference

```
        const { output: summaryResult } = await summaryPrompt(input);
        summary = summaryResult;
    }

    // Stage 3: Final contextual answer
```

```
    const { output: finalAnswer } = await answerPrompt({
        ...input,
        summary
    });

    return { answer: finalAnswer, summary };
  }
);
```

Listing 8.10: Complete Study Support Flow (src/ai/flows/ai-based-study-support.ts)

8.4 DESIGN SYSTEM AND VISUAL IDENTITY

The visual identity of Aether Co-Pilot is defined by a curated palette of midnight blues, vibrant purples, and semi-transparent "glassmorphism" effects. This is implemented using a custom Tailwind CSS configuration and a global CSS design system.

8.4.1 Tailwind CSS Configuration

The configuration extends the default theme with custom font families, brand colors, and sophisticated "blob" animations for background aesthetics.

```
import type { Config } from 'tailwindcss';

export default {
  darkMode: ['class'],
  theme: {
    extend: {
      fontFamily: {
        body: ['var(--font-inter)', 'sans-serif'],
        code: ['var(--font-roboto-mono)', 'monospace'],
      },
      colors: {
        background: 'hsl(var(--background))',
        primary: { DEFAULT: 'hsl(var(--primary))' },
        accent: { DEFAULT: 'hsl(var(--accent))' },
        brand: {
```

```
          primary: '#2C3E50',
          accent: '#8E44AD',
        }
      },
      keyframes: {
        blob: {
          '0%': { transform: 'translate(0px, 0px) scale(1)' },
          '33%': { transform: 'translate(30px, -50px) scale(1.1)' },
          '100%': { transform: 'translate(0px, 0px) scale(1)' },
        },
      },
      animation: {
        'blob': 'blob 7s infinite',
      },
    },
  },
} satisfies Config;
```

Listing 8.11: Tailwind Design Configuration (tailwind.config.ts)

8.4.2 Global CSS Tokens

The application uses HSL (Hue, Saturation, Lightness) variables to allow for easy theme switching and precise color control.

```
@layer base {
  :root {
    --background: 210 40% 98%;
    --foreground: 222 47% 11%;
    --primary: 222 47% 11%;
    --accent: 262 83% 58%;
    --radius: 0.75rem;
  }

  .dark {
    --background: 222 47% 11%;
```

```
    --foreground: 210 40% 98%;
    --accent: 263 70% 50%;
  }
}

.glass-morphism {
  background: rgba(255, 255, 255, 0.05);
  backdrop-filter: blur(10px);
  border: 1px solid rgba(255, 255, 255, 0.1);
}
```

Listing 8.12: Global CSS Design Tokens (src/app/globals.css)

8.4.3 Task Automation Flow (Natural Language to Shell)

This module allows users to automate OS-level or web tasks by describing them in plain English. The AI generates an executable script and provides a logical breakdown of its operation.

```
'use server';
import { ai } from '@/ai/genkit';
import { z } from 'genkit';

const AutomateTaskInputSchema = z.object({
  taskDescription: z.string().describe('Repetitive task to automate.
    '),
});

const AutomateTaskOutputSchema = z.object({
  automationScript: z.string().describe('The generated script.'),
  explanation: z.string().describe('Logical breakdown of the script.
    '),
});

export const automateTaskFlow = ai.defineFlow(
  {
```

```
    name: 'automateTaskFlow',
    inputSchema: AutomateTaskInputSchema,
    outputSchema: AutomateTaskOutputSchema,
  },
  async (input) => {
    try {
      const { output } = await withRetry(() => automateTaskPrompt(
   input));
      return output!;
    } catch (error: any) {
      return {
        automationScript: "# Fallback script\necho 'Please try again
    later.'",
        explanation: "The connection to the AI engine timed out."
      };
    }
  }
);
```

Listing 8.13: Task Automation Engine (src/ai/flows/task-automation.ts)

8.5 IDENTITY AND SECURITY SYNC LAYER

The bridge between our primary authentication provider (Clerk) and our database security engine (Firestore) is a specialized React hook. This layer ensures that every user session is cryptographically bound to a unique Firebase identity.

8.5.1 Identity Synchronization Hook (Clerk to Firebase)

This hook fetches a custom Firebase token from our backend API using the Clerk session and signs the user into Firebase.

```
'use client';
import { useAuth } from '@clerk/nextjs';
import { initializeApp } from 'firebase/app';
import { getAuth, signInWithCustomToken } from 'firebase/auth';
import { useEffect } from 'react';
```

```
6
7 export function useClerkFirebaseSync() {
8   const { getToken, userId } = useAuth();
9
10  useEffect(() => {
11    const syncAuth = async () => {
12      if (!userId) return;
13
14      try {
15        // Fetch the unique Firebase token generated by our server
16        const firebaseToken = await getToken({ template: '
    integration-firebase' });
17
18        if (firebaseToken) {
19           const auth = getAuth();
20           await signInWithCustomToken(auth, firebaseToken);
21           console.log('Firebase Auth Synced with Clerk UID:',
    userId);
22        }
23      } catch (error) {
24        console.error('Auth Synchronization Failed:', error);
25      }
26    };
27
28    syncAuth();
29  }, [userId, getToken]);
30 }
```

Listing 8.14: Clerk-Firebase Auth Sync Hook (src/firebase/clerk-firebase-sync.tsx)